New York, New York
cfilion@flatironinstitute.org

Carrie Filion

Research Fellow

carriefilion.github.io
github.com/CarrieFilion

Interests:
Local Group archaeology, galactic dynamics, galactic chemical evolution, spectroscopic surveys, data sonification

Summary Statistics

Publications ◇ **6** first author publications ◇ **18** supporting author publications ◇ H-index of **10** ◇ **216** citations[1]
Grants ◇ **4** successful PI/Co-PI Telescope proposals ◇ awarded **$260,264** in grants support ◇ NASA FINESST fellow
Mentorship ◇ **11** mentees ◇ **3** undergraduate mentee-led papers ◇ leadership in **3** collaborations

Appointments

Flatiron Research Fellow 2024-present
Center for Computational Astrophysics, Flatiron Institute, New York, NY

Education

The Johns Hopkins University
PhD in Astronomy and Astrophysics, 2020-2024
M.A. in Physics and Astronomy 2018-2020
Dissertation Advisor: Rosemary F.G. Wyse

Bryn Mawr College
A.B. with Honors in Physics 2015-2018
Magna Cum Laude

Awards and Grants

Instituto de Astrofísica de Canarias Early-Career visitor program 2026
Competitively awarded funds for a six week visit
American Physical Society Women in Physics Group Grants – $1,000 2023
American Astronomical Society International Travel Grant – $1,000 2022
NASA FINESST Fellowship – $150,000 2021 - 2024
JHU Alumni Association Student Grant – $1,000 2018
Note: Cannot apply for grants in current position

Awarded Telescope Time

Homogeneity in the Hyades
Proposed high resolution Magellan MIKE spectroscopic observations of a large number of likely Hyades members to constrain detailed abundances and complement our Hubble ultraviolet program Co-PI, *successful*, 2026

All That Glitters Isn't Gold (But It Could Be!): Constraining 3rd-Peak R-Process Abundances in Hyades
Proposed Hubble Space Telescope STIS spectroscopic observations of stars in the Hyades open cluster to constrain heavy element abundances. ***Awarded $107,264*** Co-PI, *successful*, 2026

Mapping of the Andromeda Satellite Galaxies with Triple Narrow-Bands
Proposed a Hyper Suprime-Cam imaging campaign of select Andromeda satellite galaxies Co-I, *successful*, 2026

Prime Focus Spectrograph Subaru Strategic Program (PFS SSP)
Proposed, as a part of a multi-national collaboration, a multi-year, 365-night survey to be conducted on the Subaru PFS. Survey is now taking data Co-I, *successful*, 2024

Cool Subdwarfs in the Galactic Halo
Proposed to observe candidate cool subdwarfs in the stellar halo using unfilled fibers on PFS. Co-PI, *successful*, 2023

Exploring the Spatial Extent of Boötes I Ultra Faint Dwarf Spheroidal Galaxy with Blue Horizontal Branch Stars

[1] stats from Google Scholar, as of 10/29/2026

New York, New York
cfilion@flatironinstitute.org

Carrie Filion

Research Fellow

carriefilion.github.io
github.com/CarrieFilion

Proposed a program to obtain spectra of candidate member stars of the Boötes I ultra faint dwarf galaxy on the Apache Point Observatory 3.5 meter PI, *successful*, 2021

The ZERO survey - Search for zero-metal stars in the Galaxy with HSC/NB395
Proposed for 2.5 nights on Subaru's Hyper Suprime-Cam to search for metal-poor stars Co-I, *successful*, 2021

Mentoring

Dissertation Advisor, Primary Mentor:
Mentor and Primary Advisor, *Masters Student*: G.G Since Fall 2025
Mentored a City University New York masters bridge program thesis on a spectroscopic outlier detection project

Project Advisor and Mentor:
Mentor and Advisor, *Masters Student*: D.O. Since Spring 2026
Advised a New York University music technology masters student on a sonification research project
Mentor and Advisor, *Ph.D. Student*: Z.H Since Spring 2026
Served as a science advisor to a Ph.D. candidate at the University of Texas on a Gaia spectroscopy project
Mentor and Advisor, *Ph.D. Student*: A.S. Fall 2025 - Fall 2026
Served as a science advisor to a Ph.D. candidate at the University of Utah on an M-dwarf spectroscopy project
Mentor and Advisor, *Undergraduate Student*: D.R. Summer 2025 - Fall 2025
Mentored an undergraduate student from North Carolina Central State through the National Society of Black Physicists program on a data driven stellar abundances project. Student is now in a masters bridge program
Mentor and Advisor, *Ph.D. Student*: S.H. Summer - Fall 2025
Served as a science and career advisor to a Ph.D. candidate at the University of Arizona
Mentor and Advisor, *Undergraduate Student*: K.D. 2022-2025
Mentored and co-advised an undergraduate student at Johns Hopkins University on a Galactic archaeology research project. Student is now in a PhD program

Co-Advisor/Co-Mentor:
Career Mentor, *Masters Student*: E.M. 2024-2026
Served as a career mentor and project advisor for a City University of New York masters bridge program student. Student is now in a PhD program
Undergraduate Students: A.G., C.L., M.H.M. 2024-2025, 2025-2026
Co-advised undergraduate students at the University of Edinburgh on basis function expansion-oriented projects. Two students are now in PhD programs

Advising Groups:
Sonification Group Since Spring 2026
Leading a group of NYU students in the Music and Audio Research Laboratory group in sonification projects
Spectra Summer Camp Summer 2025
Built an eleven person mentorship pod to support both students and mentors working on related projects

Selected Presentations

Invited:
University of Maryland Center for Theory and Computation Seminar, *College Park, Maryland* 2026
Yale Data Science x Astronomy-Astrophysics Seminar, *New Haven, Connecticut* 2026
Department of Astronomy and Space Sciences at Istanbul University Talks series, *Instanbul, Turkey* 2026
Invited talk: Bryn Mawr College colloquium, *Bryn Mawr, Pennsylvania* 2025
American Museum of Natural History Seminar, *New York, New York* 2023
University of Notre Dame Astrophysics Seminar, *South Bend, Indiana* 2022

New York, New York
cfilion@flatironinstitute.org

Carrie Filion

Research Fellow

carriefilion.github.io
github.com/CarrieFilion

University of Wisconsin-Madison Monday Science Seminar, *Madison, Wisconsin* 2022

Conference Contributions:

246th American Astronomical Society meeting, *Anchorage, AK* 2025
International Conference for High Performance Computing, *St. Louis, Missouri* 2025
National Astronomy Meeting, *Durham, UK* two talks, 2025
IAUS 377 Early Disk-Galaxy Formation from JWST to the Milky Way, *Kuala Lumpur, Malaysia* 2023

Selected Workshops and Schools

Organized:

Exoplanets in the Galactic Context, Flatiron Institute, NYC 2026
Science organizing committee, ran workshop to bridge the exoplanet and galactic astronomy communities

Sonification: from Data to Sound, Math for America, NYC 2025, 2026
Lead workshops for math and science teachers in the NYC area

Sonification for Research: Techniques, Efficacy, and Applications (SoniTEA), Flatiron Institute, NYC 2025
Lead a workshop on sonification and audio display for research applications

Basis Function Expansions for Galactic Dynamics Workshop, Tucson, AZ 2024
Science organizing committee, ran a workshop for graduate students at the University of Arizona

Additional Training:

Universe of Sound Astrophysical Data Sonification Workshop, Center for Astrophysics, Cambridge, MA 2024

Stellar Astrophysics in the Era of Gaia, Spectroscopic, and Asteroseismic Surveys MIAPbP, Garching, Germany 2023
Competitively chosen to attend

Wide-Field Spectroscopy vs Galaxy Formation Theory, Tucson, AZ 2023
Co-chaired a session on M31 and the Milky Way

First Author Publications

1. **Filion, C.** et al., 2025, "Counterculture Stars: Slow and Retrograde Stars with Low-Alpha Disk Abundances", ApJ, 989, 70
2. **Filion, C.** et al., 2024, "The Low-Mass Stellar Initial Mass Function in Nearby Ultra-Faint Dwarf Galaxies", ApJ, 967, 165
3. **Filion, C.** et al., 2023, "The Non-Axisymmetric Influence: Radius and Angle-Dependent Trends in a Barred Galaxy", MNRAS, 524, 1
4. **Filion, C.** et al., 2022, "Faint Stars in a Faint Galaxy. II. the Low-Mass Stellar Initial Mass Function of the Boötes I Ultrafaint Dwarf Galaxy", ApJ, 939, 38
5. **Filion, C.** & Wyse, R.F.G., 2021, "The Far-away Blues: Exploring the Furthest Extents of the Boötes I Ultra-faint Dwarf Galaxy", ApJ, 923, 218
6. **Filion, C.** et al., 2020, "Faint Stars in a Faint Galaxy. I. Ultradeep Photometry of the Boötes I Ultrafaint Dwarf Galaxy", ApJ, 901, 82

Additional Publications

Refereed

1. Geda, R., et al., 2026, Galactic Dynamics Drives Cyclic Star Formation in Dipole Dwarf Galaxies, *Submitted*
2. Huang, B., et al., 2026, The Zero Enrichment Rare Objects (ZERO) Survey I: Photometric Calibration of NB395 and Metallicity Estimation, ApJS, *Accepted*

New York, New York
cfilion@flatironinstitute.org

Carrie Filion

Research Fellow

carriefilion.github.io
github.com/CarrieFilion

3. Horta, D., et al. 2026, The Milky Way's Circular Velocity Curve Measured Using Element Abundance Gradients, ApJ, *1000, 8*
4. Jurado, C., et al. 2026, Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from Kinematic Ridges with Gaia, ApJ, *998, 152*
5. Rose, J.C., et al. 2026, The DREAMS Project: Disentangling the Impact of Halo-to-Halo Variance and Baryonic Feedback on Milky Way Satellite Galaxies, ApJ, *1002, 2*
6. Rose, J.C., et al. 2026, The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties, *submitted*, ApJ
7. *† Martinez, M.H., et al. 2026, Inclinations and Position Angles for Disc Galaxies in the SGA sample, RASTI, *submitted*
8. Kane, S.G., et al. 2025, On the connection between nitrogen-enhanced field stars and the Galactic globular clusters, MNRAS, *546, 1*
9. *† Ding, K., et al. 2025, Machine Learning Techniques to Distinguish Giant Stars from Dwarf Stars Using Only Photometry - Pushing Redwards, ApJ, *170, 6*
10. Manea, C., et al. 2025, Optical Spectroscopy Reveals Hidden Neutron-capture Elemental Abundance Differences among APOGEE-identified Chemical Doppelgängers, ApJ, 993, 1
11. McClure, R.L, et al. 2025, Secular Attrition of Classical Bulges by Stellar Bars, ApJ, *Submitted*
12. * Esser, N., et al., 2025, "Constraints on asteroid-mass primordial black holes in dwarf galaxies using Hubble Space Telescope photometry", A&A, 698, A290
13. † Ganapathy, A., et al., 2025, "Disc Asymmetry Characterisation in JWST-Observed Galaxies at 1 < z < 4" , MNRAS, 539, 2
14. McClure, R.L., et al., 2025, "The Impact of Classical Bulges on Stellar Bars and Boxy-Peanut-X Features in Disc Galaxies", MNRAS, 537, 2
15. * Hackshaw, Z., et al., 2024, "[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE", ApJ, 977, 143
16. Ogami, I., et al., 2024, "Detection of a Spatially Extended Stellar Population in M33: A Shallow Stellar Halo?", ApJ, 971, 107
17. Ogami, I., et al., 2024, "The structure of the stellar halo of the Andromeda galaxy explored with the NB515 for Subaru/HSC. I.: New Insights on the stellar halo up to 120 kpc", MNRAS, 536, 1
18. Hunt, J., et al., 2024, "Radial Phase Spirals in the Solar Neighbourhood", MNRAS, 527, 11393

* - Indicates second or third author, †- Indicates mentee paper

White Papers

1. Galactic Archaeology with the Subaru ʻŌnohiʻula Prime Focus Spectrograph Strategic Program — Co-I, 2026

   Co-authored a white paper detailing the Galactic archaeology component of the PFS survey, available at arXiv:2604.09875

2. RomAndromeda: The Roman Survey of the Andromeda Halo — Co-I, 2023

   Co-authored a white paper detailing a 2-passband, 2-epoch wide-field Roman imaging campaign on the halo of M31, available at arXiv:2306.12302

New York, New York
cfilion@flatironinstitute.org

Carrie Filion

Research Fellow

carriefilion.github.io
github.com/CarrieFilion

Selected Service and Community Engagement

Museum of the Moving Image public lecture	2026
Center for Computational Astrophysics 10 Year Anniversary Celebration, organizing committee	2026
Sandbox Films Summer Camp, scientist attendee at interdisciplinary film retreat	2026
Committee member for Committee for Accessibility Rights and Equity, American Astronomical Society	Since 2026
Night at the Library public talk, Brooklyn Public Library	2026
Co-producer of Labocine short-form documentary film	2025
Working Group on Accessibility and Disability, American Astronomical Society	2025
Hiring committee, Center for Computational Astrophysics	2025
Review Panelist, National Science Foundation	2025
Invited public talk, New Jersey State Museum	2025, 2026
Local organizing committee, Women in Physics Summit, JHU	2022, 2023
Co-Founder & Host, No-Z Galaxies Journal Club	2022
Journal Referee	Since 2021
Community Engagement President	2018-2021
Head Teaching Assistant, AS 171.104 General Physics for Biology Majors	2019

Collaboration Leadership Roles

EXP Development Group
Core member and code contribute to `EXP` group, developing a holistic approach to galaxy dynamical evolution using basis function expansions, see `exp-code.github.io/`

DaRk mattEr with AI and siMulationS (DREAMS) Project
Lead of the dynamics working group within the DREAMS project, working to run large numbers of cosmological simulations to better understand the nature of dark matter and galaxy formation

Prime Focus Spectrograph (PFS) Collaboration
An international collaboration dedicated to planning and carrying out a 360-night survey on the Subaru PFS. I have **earned data rights** to the galactic archaeology component of the survey